JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2026 4th April Evening Shift

Question Paper with Answer Key

75	ENGLISH	DARK
QUESTIONS	LANGUAGE	FORMAT

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Chemistry	25 questions
Mathematics	25 questions
Physics	25 questions
Answer key	Included at the end

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QR codes open the DekhoExams website.

Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

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The correct order of total number of atoms present in

(A) 2 moles of cyclohexane

(B) 684 g of sucrose

(C) 90.8 L of dihydrogen at STP

is :

(A) $C > A > B$ **(B)** $C > B > A$ **(C)** $B > C > A$ **(D)** $B > A > C$

Q2. MCQ (Single Correct Answer)

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The species having identical radii according to the Bohr's theory are :

A. H (first orbit)

B. He^{+}(first orbit)

C. He^{+}(Second orbit)

D. Li^{2+} (first orbit)

E. Be^{3+} (Second orbit)

Choose the correct answer from the options given below :

(A) A and C Only **(B)** A and E Only **(C)** B and E Only **(D)** C and D Only

Q3. MCQ (Single Correct Answer)

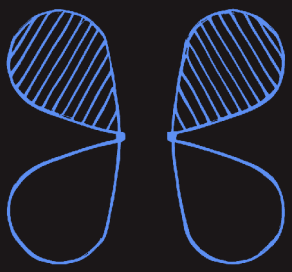
Which of the following pictorial diagrams most correctly represents the π^* (π antibonding) molecular orbital between two atoms if the internuclear axis is taken to be in the z-direction ($\xrightarrow{z\text{-axis}}$) ?

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(A)



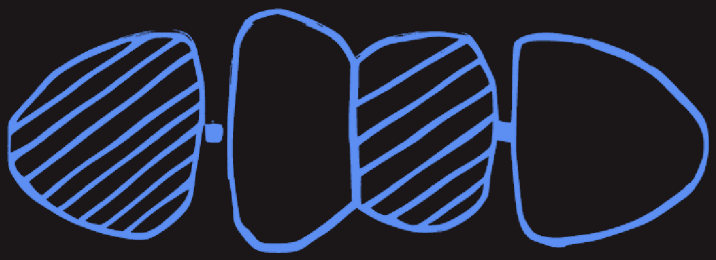
(B)



(C)

Diagram pending

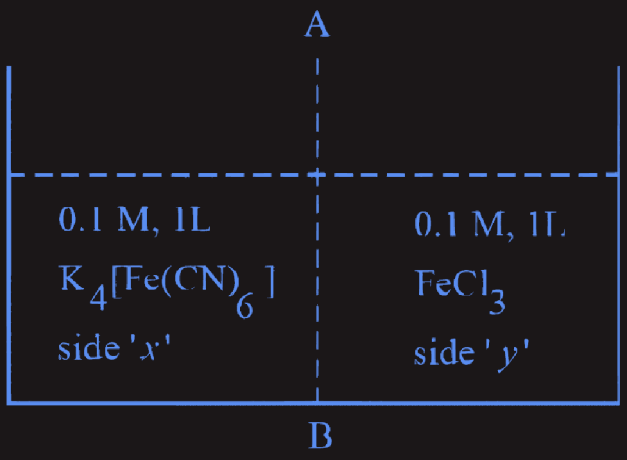
(D)



Q4. MCQ (Single Correct Answer)

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At $27^{\circ}\mathrm{C}, 0.1\mathrm{M}, 1\ \mathrm{L}\ \mathrm{K}_4\left[\mathrm{Fe(CN)}_6\right]$ aqueous solution and $0.1\mathrm{M}, 1\ \mathrm{L}\mathrm{FeCl}_3$ aqueous solution are placed in a container separated by a semi permeable membrane AB . Assume complete dissociation of both the solutes. Which of the following statement is correct?



(A) Blue color is formed on both sides.

(B) Ionic solutes in aqueous solution can pass through semi-permeable membrane.

(C) Solution on side ' y ' is hypotonic.

(D) To cause the reverse flow of solvent during osmosis, external pressure (any value) should be applied to side ' x '.

Q5. MCQ (Single Correct Answer)

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20 mL of a solution of acetic acid required 28.4 mL of 0.1 M NaOH for its neutralization. A solution (X) was prepared by mixing 20 mL of the above acetic acid and 14.2 mL of 0.1 M NaOH solution. What is the pH of the solution (X) ? $\left(\mathrm{pK}_{\mathrm{a}}\right.$ value of acetic acid is 4.75).

(A) 7.0 (B) 4.75 (C) 3.5 (D) 4.82

Q6. MCQ (Single Correct Answer)

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Match the LIST-I with LIST-II

List-I Reaction		List-II Mechanism	
A.	Williamson Synthesis	I.	Electrophilic addition
B.	Friedel Craft Reaction	II.	Free radical substitution
C.	Bromination of vinyl benzene	III.	Nucleophilic substitution
D.	Chlorination of toluene in light	IV.	Electrophilic substitution

Choose the correct answer from the options given below:

(A) A-III, B-I, C-II, D-IV

(B) A-III, B-IV, C-II, D-I

(C) A-III, B-IV, C-I, D-II

(D) A-I, B-III, C-IV, D-II

Q7. MCQ (Single Correct Answer)

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The 1st ionization enthalpy for Mg is $+737\ \mathrm{kJ/mol}$. The most probable estimated value of the 2nd ionization enthalpy of Mg is ____ .

(A) $-906\ \mathrm{kJ/mol}$

(B) $-856\ \mathrm{kJ/mol}$

(C) $+1450\ \mathrm{kJ/mol}$

(D) $+590\ \mathrm{kJ/mol}$

Q8. MCQ (Single Correct Answer)

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The electronegativity of a group 13 element ' E ' is same as that of Ge (on Pauling scale and upto one decimal point). The CORRECT statements about E^{3+} are

A. It can act as a reducing agent.

B. It can act as an oxidizing agent.

C. E^{3+} is more stable than E^{+}.

D. The standard electrode potential value for $\mathrm{E}^{3+}/\mathrm{E}$ is positive.

Choose the correct answer from the options given below :

(A) A and C Only

(B) B and C Only

(C) B and D Only

(D) A and D Only

Q9. MCQ (Single Correct Answer)

Pairs of elements with the same number of electrons in their respective 4f orbital are

[Atomic number. Eu-63, Gd-64, Dy-66, Ho-67, Tm-69, Yb-70, Lu-71, Hf-72]

A. (Eu and Gd)

B. (Dy and Ho)

C. (Yb and Hf)

D. (Lu and Tm)

Choose the correct answer from the options given below :

(A) B and C Only (B) A and B Only (C) A and D Only (D) A and C Only

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Q10. MCQ (Single Correct Answer)

Consider the metal complexes $[\mathrm{Ni(en)_3}]^{2+}(\mathrm{A})$, $[\mathrm{NiCl_4}]^{2-}(\mathrm{B})$ and $[\mathrm{Ni(NH_3)_6}]^{2+}(\mathrm{C})$. Choose the CORRECT option by considering the number of unpaired electrons present in (A), (B) and (C) respectively and the order of frequency of absorption.

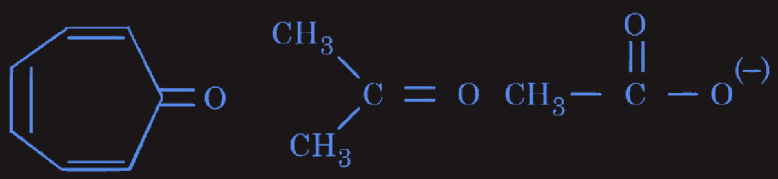
(A) 2, 2, 2 and $(\mathrm{A}) > (\mathrm{C}) > (\mathrm{B})$ (B) 0, 2, 0 and $(\mathrm{A}) > (\mathrm{C}) > (\mathrm{B})$

(C) 2, 2, 0 and $(\mathrm{B}) > (\mathrm{C}) > (\mathrm{A})$ (D) 2, 2, 2 and $(\mathrm{C}) > (\mathrm{A}) > (\mathrm{B})$

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Q11. MCQ (Single Correct Answer)

Consider the following molecules/species :



(x) (y) (z)

The correct order of carbon-oxygen double bond length is :

(A) $x > y > z$ (B) $\mathrm{y} > \mathrm{z} > x$ (C) $\mathrm{z} > x > \mathrm{y}$ (D) $x > \mathrm{z} > \mathrm{y}$

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Q12. MCQ (Single Correct Answer)

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Consider $|x|$ is the difference in oxidation states of Mn in highest manganese fluoride and highest manganese oxide. The ions with $|x|$ number of unpaired electrons from the following are :

A. Sc^{3+}

B. Zn^{2+}

C. V^{2+}

D. Fe^{2+}

E. Co^{2+}

Choose the correct answer from the options given below :

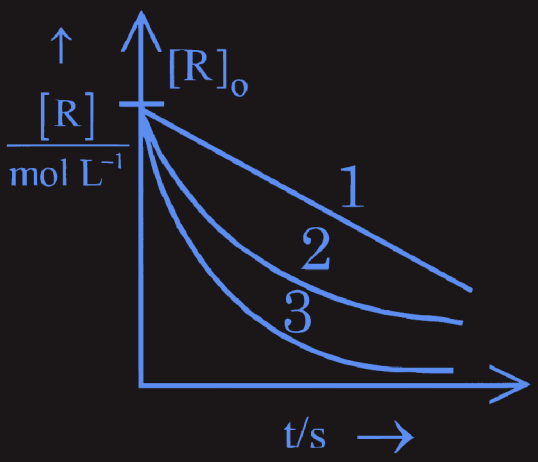
(A) A and B Only (B) C, D and E Only (C) C and E Only (D) B and E Only

Q13. MCQ (Single Correct Answer)

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Consider the given graph showing variation of reactant concentration with time.

Three different reactions were started with identical initial concentration of reactants. Which of the following statement is correct?



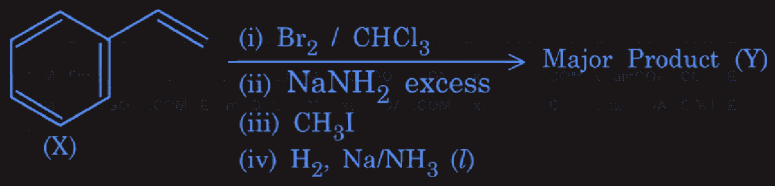
(A) The order of all the three reactions is same.

(B) The rate constant of reaction 3 is larger than the rate constant of reaction 2 if the order of reaction is same for both.

(C) The SI unit of rate constant of reaction 1 is s^{-1}.

(D) Thermal decomposition of HI on gold surface is an example of reaction 2 .

Q14. MCQ (Single Correct Answer)



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Compound (X) is subjected to the sequence of reactions as shown above. Molar mass of the major product (Y) formed is ____ $gmol^{-1}$.

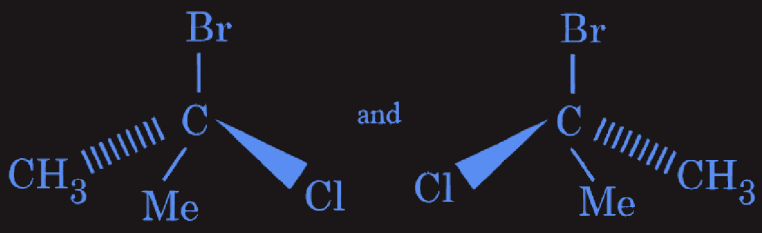
(Given molar mass in $gmol^{-1}$ C : 12, H : 1, O : 16)

(A) 90 (B) 118 (C) 160 (D) 125

Q15. MCQ (Single Correct Answer)

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The following structures are :

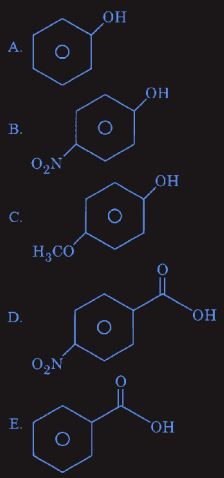


(A) enantiomers.
(B) identical molecules.
(C) diastereomers.
(D) meso compounds.

Q16. MCQ (Single Correct Answer)

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The descending order of acidity among the following compounds is :



Choose the correct answer from the options given below :

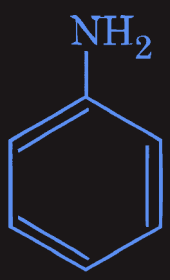
(A) $B > D > E > A > C$
(B) $D > B > E > A > C$
(C) $C > A > B > D > E$
(D) $D > E > B > A > C$

Q17. MCQ (Single Correct Answer)

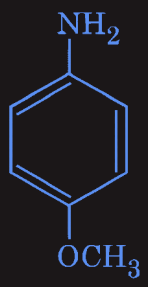
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The strongest conjugate acid will result from :

(A)



(B)



(C)



(D)

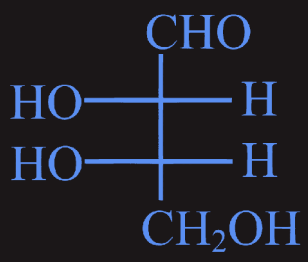


Q18. MCQ (Single Correct Answer)

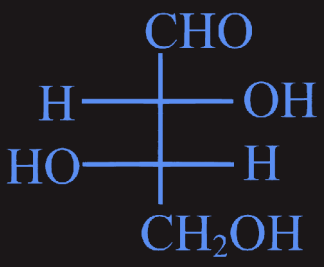
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A D-aldotetrose on oxidation with concentrated HNO_3 resulted in optically inactive dicarboxylic acid. The structure of the D-aldotetrose is :

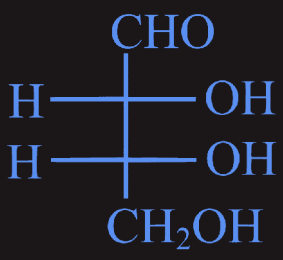
(A)



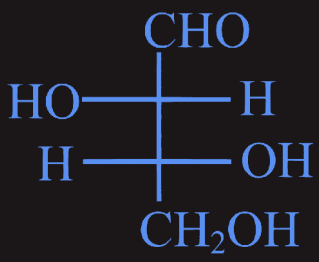
(B)



(C)



(D)



Q19. MCQ (Single Correct Answer)

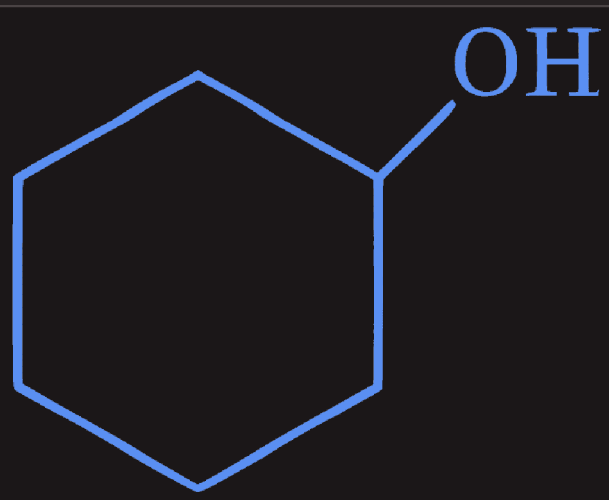
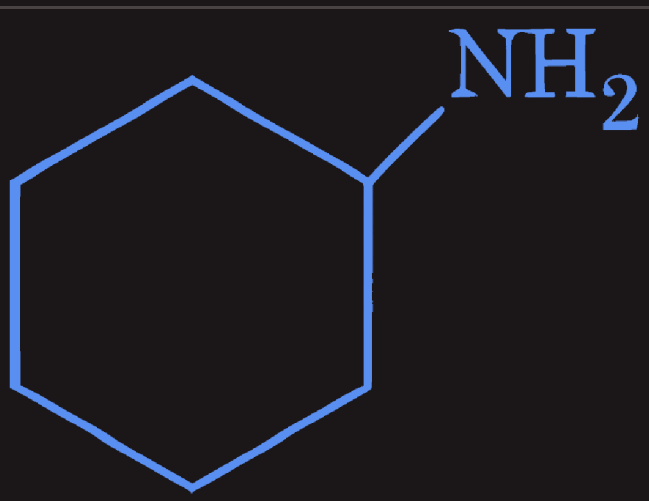
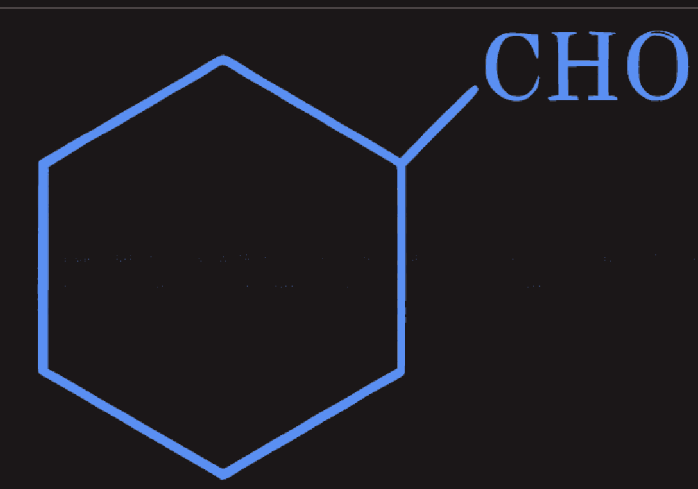
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Among Fe^{3+}, Pb^{2+}, Cu^{2+} and Mn^{2+}, identify the one that gets precipitated out while passing H_2S in presence of NH_4OH as group reagent. The highest possible oxidation state of the corresponding metal is

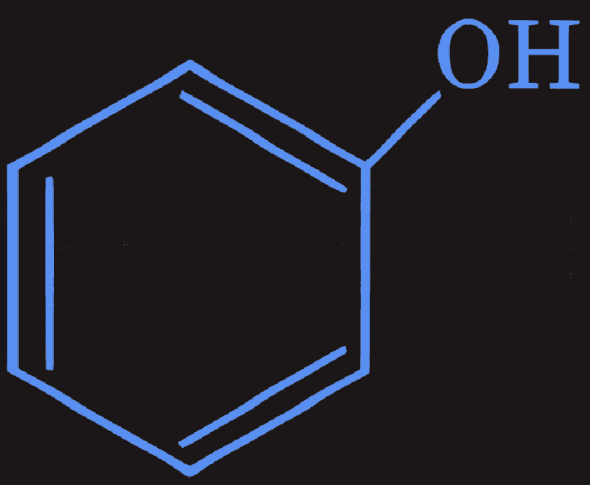
(A) +3 (B) +4 (C) +2 (D) +7

Q20. MCQ (Single Correct Answer)

Match the LIST-I with LIST-II

List - I		List - II	
	Compound		**Test**
A.		I.	Hinsberg's reagent test
B.		II.	Phthalein dye test
C.		III.	Lucas test

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D.		IV.	Tollen's test

Choose the correct answer from the options given below :

(A) A-III, B-I, C-IV, D-II

(B) A-III, B-IV, C-I, D-II

(C) A-I, B-III, C-II, D-IV

(D) A-I, B-II, C-III, D-IV

Q21. Numerical

If 3.365 g of ethanol (l) is burnt completely in a bomb calorimeter at 298.15 K , the heat produced is 99.472 kJ . The $|\Delta \mathrm{H_f}^\circ|$ of ethanol at 298.15 K is

____ $\times 10^2\ \mathrm{kJ\ mol^{-1}}$. (Nearest integer)

Given: Standard enthalpy for combustion of graphite $= -393.5\ \mathrm{kJ\ mol^{-1}}$

Standard enthalpy of formation of water $(\mathrm{l}) = -285.8\ \mathrm{kJ\ mol^{-1}}$

Molar mass in $\mathrm{gmol^{-1}}$ of $\mathrm{C, H, O}$ are 12,1 and 16 respectively

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Q22. Numerical

For the following reaction at $50^\circ\mathrm{C}$ and at 2 atm pressure,

$2\,\mathrm{N_2O_5(\,g)} \rightleftharpoons 2\,\mathrm{N_2O_4(\,g)} + \mathrm{O_2(\,g)}$

$\mathrm{N_2O_5}$ is 50% dissociated.

The magnitude of standard free energy change at this temperature is x.

$x =$ ____ $\mathrm{Jmol^{-1}}$ [Nearest integer].

Given : $\mathrm{R} = 8.314\ \mathrm{J\ mol^{-1}\ K^{-1}}, \log 2 = 0.30, \log 3 = 0.48, \ln 10 = 2.303, {}^\circ\mathrm{C} + 273 = \mathrm{K}$

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Q23. Numerical

An electrochemical cell consists of the following two redox couples, $M^{x+}(aq)/M(s)\left[E^{\Theta}_{red} = +0.15\ V\right]$ and $Fe^{3+}(aq)/Fe(s)\left[E^{\Theta}_{red} = -0.036\ V\right]$. The cell EMF (E_{cell}) is recorded to be 0.2057 V . If the reaction quotient of the electrochemical reaction is found to be 10^{-2}, then the value of x is

____ .(Nearest integer)

[Given : M is a p-block metal and $\frac{2.303RT}{F} = 0.059\ V$]

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Q24. Numerical

For a first order reaction $A \rightarrow B$

$t/\min$	$[A]/M$
0	0.6500
x	0.0650
20	0.00065

$x =$ ____ min. (Nearest integer)

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Q25. Numerical

In sulphur estimation, 2.0×10^{-3} mol of an organic compound (X) (molar mass $76\ g\ mol^{-1}$) gave 0.4813 g of barium sulphate (molar mass $233\ g\ mol^{-1}$). The percentage of sulphur in the compound (X) is ____ $\times 10^{-1}\%$ (Nearest integer)

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

For the function $f : [1, \infty) \rightarrow [1, \infty)$ defined by $f(x) = (x-1)^4 + 1$, among the two statements:

(I) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x)\}$ contains exactly two elements, and

(II) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x+1)\}$ is an empty set,

(A) only (I) is TRUE

(B) only (II) is TRUE

(C) both (I) and (II) are TRUE

(D) neither (I) nor (II) is TRUE

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Q27. MCQ (Single Correct Answer)

Let $S = \{z \in C : z^2 + 4z + 16 = 0\}$. Then $\sum_{z \in S} |z + \sqrt{3}i|^2$ is equal to :

(A) 42

(B) 23

(C) 27

(D) 38

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Q28. MCQ (Single Correct Answer)

If the system of equations :

$x + y + z = 5$
$x + 2y + 3z = 9$
$x + 3y + \lambda z = \mu$

has infinitely many solutions, then the value of $\lambda + \mu$ is :

(A) 16 (B) 18 (C) 19 (D) 21

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Q29. MCQ (Single Correct Answer)

If $\alpha = 1$ and $\beta = 1 + i\sqrt{2}$, where $i = \sqrt{-1}$ are two roots of the equation $x^3 + ax^2 + bx + c = 0, a, b, \in \mathbb{R}$, then $\int_{-1}^{1} (x^3 + ax^2 + bx + c)dx$ is equal to:

(A) -2 (B) -4 (C) -8 (D) -10

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Q30. MCQ (Single Correct Answer)

If the quadratic equation $(\lambda + 2)x^2 - 3\lambda x + 4\lambda = 0, \lambda \neq -2$, has two positive roots, then the number of possible integral values of λ is :

(A) 1 (B) 2 (C) 3 (D) 4

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Q31. MCQ (Single Correct Answer)

Let $A = \begin{bmatrix} 1 & 2 & 7 \\ 4 & -2 & 8 \\ 3 & 8 & -7 \end{bmatrix}$ and $\det(A - \alpha I) = 0$, where α is a real number. If the largest possible value of α is p, then the circle $(x - p)^2 + (y - 2p)^2 = 320$, intersects the co-ordinate axes at

(A) 1 point (B) 2 points (C) 3 points (D) 4 points

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Q32. MCQ (Single Correct Answer)

Let $\alpha = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots \infty$ and

$\beta = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \ldots \infty$. Then the value of

$(0.2)^{\log_{\sqrt{5}}(\alpha)} + (0.04)^{\log_5(\beta)}$ is equal to :

(A) 4 (B) 5 (C) 8 (D) 25

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Q33. MCQ (Single Correct Answer)

For 10 observations $x_1, x_2, \ldots, x_{10}$, if $\sum_{i=1}^{10} (x_i + 2)^2 = 180$ and $\sum_{i=1}^{10} (x_i - 1)^2 = 90$, then their standard deviation is :

(A) 2 (B) $\sqrt{3}$ (C) $2\sqrt{2}$ (D) 3

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Q34. MCQ (Single Correct Answer)

In the expansion of $\left(9x - \frac{1}{3\sqrt{x}}\right)^{18}, x > 0$, if the term independent of x is (221)k, then k is equal to:

(A) 84 (B) 78 (C) 168 (D) 198

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Q35. MCQ (Single Correct Answer)

Let $\mathrm{P}(3\cos\alpha, 2\sin\alpha), \alpha \neq 0$, be a point on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$, Q be a point on the circle $x^2 + y^2 - 14x - 14y + 82 = 0$ and R be a point on the line $x + y = 5$ such that the centroid of the triangle PQR is $\left(2 + \cos\alpha, 3 + \frac{2}{3}\sin\alpha\right)$. Then the sum of the ordinates of all possible points R is:

(A) 6 (B) 2 (C) 4 (D) 8

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Q36. MCQ (Single Correct Answer)

Let $\mathrm{H}: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be a hyperbola such that the distance between its foci is 6 and the distance between its directrices is $\frac{8}{3}$. If the line $x = \alpha$ intersects the hyperbola H at the points A and B such that the area of the triangle AOB is $4\sqrt{15}$, where O is the origin, then α^2 equals

(A) 12 (B) 16 (C) 24 (D) 25

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Q37. MCQ (Single Correct Answer)

$\max\limits_{0 \le x \le \pi} \left(16\sin\left(\frac{x}{2}\right)\cos^3\left(\frac{x}{2}\right)\right)$ is equal to :

(A) $\frac{3\sqrt{3}}{2}$ (B) $3\sqrt{3}$ (C) $4\sqrt{3}$ (D) $6\sqrt{3}$

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Q38. MCQ (Single Correct Answer)

The shortest distance between the lines

$\vec{r} = \left(\frac{1}{3}\hat{\mathrm{i}} + 2\hat{\mathrm{j}} + \frac{8}{3}\hat{\mathrm{k}}\right) + \lambda(2\hat{\mathrm{i}} - 5\hat{\mathrm{j}} + 6\hat{\mathrm{k}})$

and $\vec{r} = \left(-\frac{2}{3}\hat{\mathrm{i}} - \frac{1}{3}\hat{\mathrm{k}}\right) + \mu(\hat{\mathrm{j}} - \hat{\mathrm{k}}), \lambda, \mu \in \mathbb{R}$, is:

(A) $\sqrt{5}$ (B) 3 (C) $2\sqrt{3}$ (D) $\sqrt{15}$

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Q39. MCQ (Single Correct Answer)

If $\left(2\alpha + 1, \alpha^2 - 3\alpha, \frac{\alpha - 1}{2}\right)$ is the image of $(\alpha, 2\alpha, 1)$ in the line $\frac{x-2}{3} = \frac{y-1}{2} = \frac{z}{1}$, then the possible value(s) of α is (are)

(A) Only 3 (B) Only 3 and - 1

(C) Only 3, $\frac{1}{4}$ and -1 (D) Only 3 and $\frac{1}{4}$

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Q40. MCQ (Single Correct Answer)

Let $\hat{u}$ and $\hat{v}$ be unit vectors inclined at an acute angle such that $|\hat{u} \times \hat{v}| = \frac{\sqrt{3}}{2}$. If $\overrightarrow{A} = \lambda\hat{u} + \hat{v} + (\hat{u} \times \hat{v})$, then λ is equal to:

(A) $\frac{4}{3}(\overrightarrow{A} \cdot \hat{u}) - \frac{2}{3}(\overrightarrow{A} \cdot \hat{v})$

(B) $\frac{2}{3}(\overrightarrow{A} \cdot \hat{u}) - \frac{1}{3}(\overrightarrow{A} \cdot \hat{v})$

(C) $\frac{4}{3}(\overrightarrow{A} \cdot \hat{u}) + \frac{2}{3}(\overrightarrow{A} \cdot \hat{v})$

(D) $(\overrightarrow{A} \cdot \hat{u}) - \frac{1}{2}(\overrightarrow{A} \cdot \hat{v})$

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Q41. MCQ (Single Correct Answer)

Let for some $\alpha \in \mathbb{R}$, $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying $f(x + y) = f(x) + 2y^2 + y + \alpha xy$ for all $x, y \in \mathbb{R}$. If $f(0) = -1$ and $f(1) = 2$, then the value of $\sum_{n=1}^{5}(\alpha + f(n))$ is :

(A) 110 (B) 140 (C) 150 (D) 170

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Q42. MCQ (Single Correct Answer)

Let

$A = \{(a, b, c) : a, b, c$ are non-negative integers and $a + b + 2c = 22\}$.

Then $n(A)$ is equal to :

(A) 121 (B) 124 (C) 144 (D) 169

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Q43. MCQ (Single Correct Answer)

The area of the region bounded by the curves $x + 3y^2 = 0$ and $x + 4y^2 = 1$ is equal to :

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) $\frac{5}{3}$

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Q44. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution of the differential equation :

$\frac{dy}{dx} + \left(\frac{6x^2+(3x^2+2x^3+4)e^{-2x}}{(x^3+2)(2+e^{-2x})}\right)y = 2 + e^{-2x}$

$x \in (-1, 2)$, satisfying $y(0) = \frac{3}{2}$. If $y(1) = \alpha\left(2 + e^{-2}\right)$, then α is equal to :

(A) $\frac{13}{8}$ (B) $\frac{6}{13}$ (C) $\frac{12}{13}$ (D) $\frac{13}{12}$

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Q45. MCQ (Single Correct Answer)

The integral $\int_0^1 \cot^{-1}\left(1+x+x^2\right)dx$ is equal to :

(A) $2\tan^{-1}2+\frac{1}{2}\log_e\left(\frac{5}{4}\right)+\frac{\pi}{2}$

(B) $2\tan^{-1}2+\frac{1}{2}\log_e\left(\frac{5}{4}\right)-\frac{\pi}{2}$

(C) $2\tan^{-1}2-\frac{1}{2}\log_e\left(\frac{5}{4}\right)+\frac{\pi}{2}$

(D) $2\tan^{-1}2-\frac{1}{2}\log_e\left(\frac{5}{4}\right)-\frac{\pi}{2}$

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Q46. Numerical

From a month of 31 days, 3 different dates are selected at random. If the probability that these dates are in an increasing A.P. is equal to $\frac{a}{b}$, where $a, b \in$ N and $\gcd(a,b)=1$, then $a+b$ is equal to ____

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Q47. Numerical

Let $f(x)=\begin{cases} e^{x-1} & ,x<0 \\ x^2-5x+6 & ,x\geq 0 \end{cases}$ and $g(x)=f(|x|)+|f(x)|$. If the number of points where g is not continuous and is not differentiable are α and β respectively, then $\alpha+\beta$ is equal to ____

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Q48. Numerical

Let A, B be points on the two half-lines $x-\sqrt{3}|y|=\alpha, \alpha>0$ at a distance of α from their point of intersection P. The line segment AB meets the angle bisector of the given half-lines at the point Q. If $PQ=\frac{9}{2}$ and R is the radius of the circumcircle of $\triangle \mathrm{PAB}$, then $\frac{\alpha^2}{R}$ is equal to ____

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Q49. Numerical

Let A, B and C be the vertices of a variable right angled triangle inscribed in the parabola $y^2=16x$. Let the vertex B containing the right angle be $(4,8)$ and the locus of the centroid of $\triangle ABC$ be a conic C_0. Then three times the length of latus rectum of C_0 is ____

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Q50. Numerical

Let f be a twice differentiable function such that

$f(x)=\int_0^x \tan(t-x)dt-\int_0^x f(t)\tan t dt, x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

Then $f''\left(\frac{\pi}{6}\right)+12f'\left(-\frac{\pi}{6}\right)+f\left(\frac{\pi}{6}\right)$ is equal to _____.

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Physics 25 questions

Q51. MCQ (Single Correct Answer)

Match the LIST-I with LIST-II

List - I		List - II	
A.	Planck's constant	I.	$\mathrm{ML^2\,T^{-2}}$
B.	Stopping potential	II.	$\mathrm{T^{-1}}$
C.	Work function	III.	$\mathrm{ML^2\,T^{-1}}$
D.	Threshold frequency	IV.	$\mathrm{ML^2\,T^{-3}\,A^{-1}}$

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II

(B) A-I, B-II, C-III, D-IV

(C) A-IV, B-III, C-I, D-II

(D) A-I, B-IV, C-III, D-II

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Q52. MCQ (Single Correct Answer)

Two cars A and B are moving in the same direction along a straight line with speeds $100\ \mathrm{km/h}$ and $80\ \mathrm{km/h}$, respectively such that car A is moving ahead of car B. A person in car B throws a stone with a speed v so that it hits the car A with a speed of $5\ \mathrm{m/s}$. The value of v is ____ $\mathrm{km/h}$.

(A) 18

(B) 28

(C) 38

(D) 48

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Q53. MCQ (Single Correct Answer)

At $t = 0$, a body of mass 100 g starts moving under the influence of a force $(5\hat{\mathrm{i}} + 10\hat{\mathrm{j}})\mathrm{N}$. After 2 s its position is $(2x\hat{\mathrm{i}} + 5y\hat{\mathrm{j}})\mathrm{m}$. The ratio $x : y$ is ____ .

(A) $1 : 2$

(B) $2 : 5$

(C) $5 : 2$

(D) $5 : 4$

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Q54. MCQ (Single Correct Answer)

If x and y coordinates of a projectile as a function of time (t) are given as $24t$ and $43.6t - 4.9t^2$, respectively, then the angle (in degrees) made by the projectile with horizontal when $t = 2\ \mathrm{s}$ is ____ .

(A) 60

(B) 45

(C) 30

(D) 75

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Q55. MCQ (Single Correct Answer)

The height in terms of radius of the earth (R), at which the acceleration due to gravity becomes $\frac{g}{9}$, where g is acceleration due to gravity on earth's surface, is ____ .

(A) $\sqrt{3}R$

(B) $2\sqrt{2}R$

(C) $2R$

(D) $\frac{4}{9}R$

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Q56. MCQ (Single Correct Answer)

A metal string A is suspended from a rigid support and its free end is attached to a block of mass M. Second block having mass 2 M is suspended at the bottom of the first block using a string B. The area of cross sections of strings A and B are same. The ratio of lengths of strings of A to B is 2 and the ratio of their Young's moduli (Y_A/Y_B) is 0.5 . The ratio of elongations in A to B is ____ .

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(A) 1 (B) 4 (C) 8 (D) 6

Q57. MCQ (Single Correct Answer)

A water spray gun is attached to a hose of cross sectional area $30\ \mathrm{cm}^2$. The gun comprises of 10 perforations each of cross sectional area of $15\ \mathrm{mm}^2$. If the water flows in the hose with the speed of $50\ \mathrm{cm/s}$, calculate the speed at which the water flows out from each perforation. (Neglect any edge effects)

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(A) $100\ \mathrm{m/s}$ (B) $10\ \mathrm{m/s}$

(C) $1000\ \mathrm{m/s}$ (D) $15 \times 10^2\ \mathrm{m/s}$

Q58. MCQ (Single Correct Answer)

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : If the average kinetic energy of $\mathrm{H_2}$ and $\mathrm{O_2}$ molecules, kept in two different sized containers are same, then their temperatures will be same.

Reason R : The r.m.s speed of $\mathrm{H_2}$ and $\mathrm{O_2}$ molecules are same at same temperature.

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Choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is true but $\mathbf{R}$ is false

(D) A is false but $\mathbf{R}$ is true

Q59. MCQ (Single Correct Answer)

The temperature of a metal strip having coefficient of linear expansion α is increased from T_1 to T_2 resulting in increase of its length by ΔL_1. The temperature is further increased from T_2 to T_3 such that the increase in its length is ΔL_2.

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Given $T_3 + T_1 = 2T_2$ and $T_2 - T_1 = \Delta T$, the value of ΔL_2 is ____ .

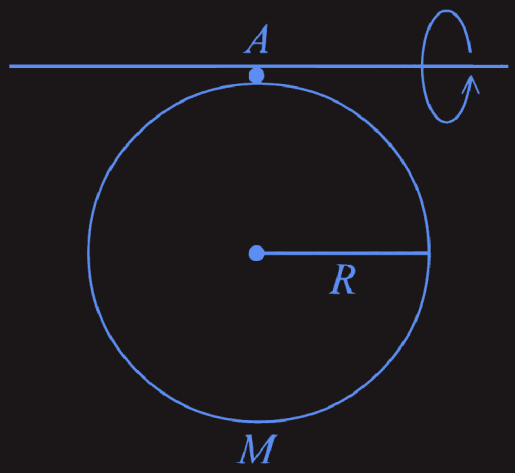
(A) $\Delta L_1 \left[1 + 2\alpha^2(\Delta T)^2\right]$ (B) $\Delta L_1 \left[1 + \alpha^2(\Delta T)^2\right]$

(C) $\Delta L_1[1 + 2\alpha\Delta T]$ (D) $\Delta L_1[1 + \alpha\Delta T]$

Q60. MCQ (Single Correct Answer)

A uniform disc of radius R and mass M is free to oscillate about the axis A as shown in the figure. For small oscillations the time period is ____ .

(g is acceleration due to gravity)



(A) $2\pi\sqrt{\frac{5R}{4g}}$ (B) $2\pi\sqrt{\frac{2R}{3g}}$ (C) $2\pi\sqrt{\frac{3R}{2g}}$ (D) $2\pi\sqrt{\frac{3R}{g}}$

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Q61. MCQ (Single Correct Answer)

A rigid dipole undergoes a simple harmonic motion about its centre in the presence of an electric field $\vec{E}_1 = E_0\hat{x}$. If another electric field $\vec{E}_2 = 2E_0(\hat{y}+\hat{z})$ is introduced to the system, what will be the percentage change in the frequency of the oscillation (approximate)?

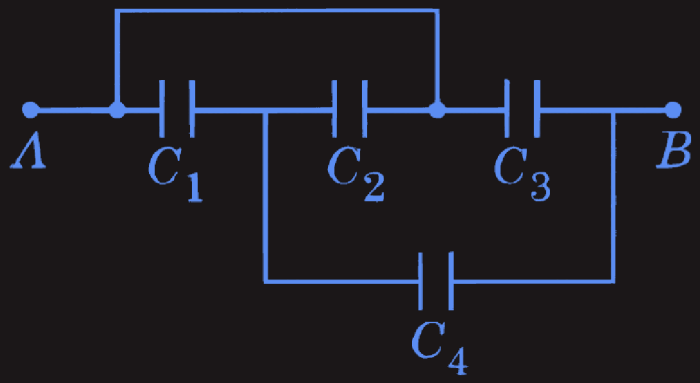
(A) 73% (B) 63% (C) 83% (D) 53%

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Q62. MCQ (Single Correct Answer)

From the circuit given below, the capacitance between terminals A and B shown in the circuit is ____ μF.

(take $C_1 = C_2 = C_3 = 1\mu\,\text{F}$ and $C_4 = 2\mu\,\text{F}$.)



(A) 2 (B) 7/2 (C) 7/3 (D) 5/2

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Q63. MCQ (Single Correct Answer)

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason $\mathbf{R}$

Assertion A : In electrostatics, a conductor does not store any net charge inside.

Reason R : Inside the capacitor (with no dielectric medium), the free charge carriers, if placed between the plates of capacitor, experience force and drift.

Choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is true but $\mathbf{R}$ is false

(D) A is false but $\mathbf{R}$ is true

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Q64. MCQ (Single Correct Answer)

A solenoid has a core made of material with relative permeability 400 . The magnetic field produced in the interior of solenoid is 1.0 T . The magnetic intensity in SI units is $\alpha \times 10^5$. The value of α is ____ .

(Free space permeability $\mu_0 = 4\pi \times 10^{-7}$ SI units.)

(A) $\frac{25}{\pi}$ (B) $\frac{1}{16\pi}$ (C) $\frac{1}{\pi}$ (D) $\frac{1}{4\pi}$

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Q65. MCQ (Single Correct Answer)

A magnetic field vector in an electromagnetic wave is represented by $\vec{B} = B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{j}$. Its associated electric field vector is ____ .

(A) $\vec{E} = -v\lambda B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{k}$

(B) $\vec{E} = -v\lambda B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{i}$

(C) $\vec{E} = v\lambda B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{k}$

(D) $\vec{E} = v\lambda B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{i}$

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Q66. MCQ (Single Correct Answer)

A convex lens is made from glass material having refractive index of 1.4 with same radius of curvature on both sides. The ratio of its focal length and radius of curvature is ____ .

(A) 0.5 (B) 2.5 (C) 0.8 (D) 1.25

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Q67. MCQ (Single Correct Answer)

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An unpolarized light of certain intensity passes through a combination of two polarizers whose transmission axes are at 30° and 90°, respectively, with respect to the horizontal axis. A third polarizer with its transmission axis at 60° with the horizontal axis is placed between the two existing polarizers. The ratio of the output intensities with and without the third polarizer is ____ .

(A) 3/4 (B) 4/3 (C) 9/4 (D) 4/9

Q68. MCQ (Single Correct Answer)

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In Rutherford's alpha-particle scattering experiment, only a few alpha particles rebound back because

A. The size of gold nucleus is very small as compared to the size of gold atom.

B. Alpha particle and gold nucleus have equal charge.

C. The impact parameter is minimum for a few alpha particles.

D. A few alpha particles have very high kinetic energy.

E. Only a few alpha particles undergo head-on collision with the nuclei.

Choose the correct answer from the options given below :

(A) A, B Only (B) B, E Only (C) C, D Only (D) A, C, E Only

Q69. MCQ (Single Correct Answer)

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The de Broglie wavelength associated with an electron accelerated through a potential difference V is λ_e and the de Broglie wavelength associated with a proton accelerated through the same potential difference is λ_p. If their corresponding masses are m_e and m_p, respectively, then the ratio of their de Broglie wavelengths $\left(\frac{\lambda_e}{\lambda_p}\right)$ is ____ .

(A) $\sqrt{\frac{m_p}{m_e}}$ (B) $\sqrt{\frac{m_e}{m_p}}$ (C) $\frac{m_p}{m_e}$ (D) $\left(\frac{m_p}{m_e}\right)^2$

Q70. MCQ (Single Correct Answer)

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Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason $\mathbf{R}$

Assertion A : A diode under reverse-biased condition provides very small current which is nearly independent of voltage until a critical limit at which the current increases drastically.

Reason R : Below the critical voltage limit, only majority charge carriers flow which increases drastically above critical voltage.

choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is true but $\mathbf{R}$ is false

(D) A is false but $\mathbf{R}$ is true

Q71. Numerical

A diode has Zener voltage of 10 V and maximum power dissipation of 0.5 W , then the minimum resistance to be used in series with this diode for safety when it is connected to a 25 V power supply is ____ Ω.

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Q72. Numerical

A gun mounted on the ground fires bullets in all directions with same speed. The farthest distance the bullets could reach is 6.4 m . The speed of the bullets from the gun is ____ $\mathrm{m/s}$.

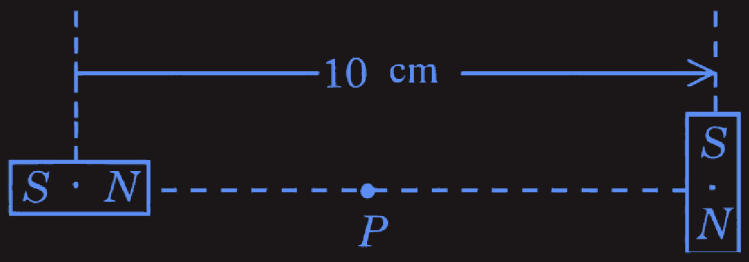
(take $\mathrm{g} = 10\,\mathrm{m/s^2}$)

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Q73. Numerical

Two identical small bar magnets each of dipole moment $3\sqrt{5}\,\mathrm{J/T}$ are placed at a center to center separation of 10 cm , with their axes perpendicular to each other as shown in figure. The value of magnetic field at the point P midway between the magnets is $\alpha \times 10^{-3}\,\mathrm{T}$. The value of α is ____

$\left(\mu_0 = 4\pi \times 10^{-7}\mathrm{Tm/A}\right)$



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Q74. Numerical

A circular coil of radius 2 cm and 125 turns carries a current of 1 A . The coil is placed in a uniform magnetic field of magnitude 0.4 T . The axis of the coil makes an angle of 30° with the direction of the magnetic field. The torque acting on the coil is $\alpha \times 10^{-4}\,\mathrm{N.m}$. The value of α is ____ .

$(\pi = 3.14)$

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Q75. Numerical

In a double slit experiment, when one of the slits is covered by a transparent mica sheet of refractive index 1.56 , the central fringe shifts to the position of 7th bright fringe, obtained with both slits uncovered. If the light source wavelength is 450 nm , the thickness of mica sheet is $\alpha \times 10^{-9}\,\mathrm{m}$. The value of α is

____ .

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Answer Key

Chemistry

1. **D**	2. **B**	3. **C**	4. **C**	5. **B**
6. **C**	7. **C**	8. **C**	9. **D**	10. **A**
11. **D**	12. **C**	13. **B**	14. **B**	15. **B**
16. **D**	17. **C**	18. **C**	19. **D**	20. **A**
21. **3**	22. **2474**	23. **2**	24. **7**	25. **435**

Mathematics

26. **A**	27. **D**	28. **B**	29. **C**	30. **B**
31. **C**	32. **C**	33. **D**	34. **A**	35. **D**
36. **B**	37. **B**	38. **B**	39. **A**	40. **A**
41. **B**	42. **C**	43. **C**	44. **D**	45. **D**
46. **944**	47. **4**	48. **9**	49. **16**	50. **5**

Physics

51. **A**	52. **C**	53. **D**	54. **B**	55. **C**
56. **D**	57. **B**	58. **C**	59. **D**	60. **A**
61. **A**	62. **A**	63. **B**	64. **B**	65. **A**
66. **D**	67. **C**	68. **D**	69. **A**	70. **C**
71. **300**	72. **8**	73. **12**	74. **314**	75. **5625**

For numerical answers given as a range, any value within the range is correct.

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